

Astrophotography *for beginners*



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Introduction & Basics



Camera's I have known and loved...

- Started in 1994... using film!
- Got my first DSLR in 2003
- Currently have a Canon 5D Mkii
- I've never owned a telescope!



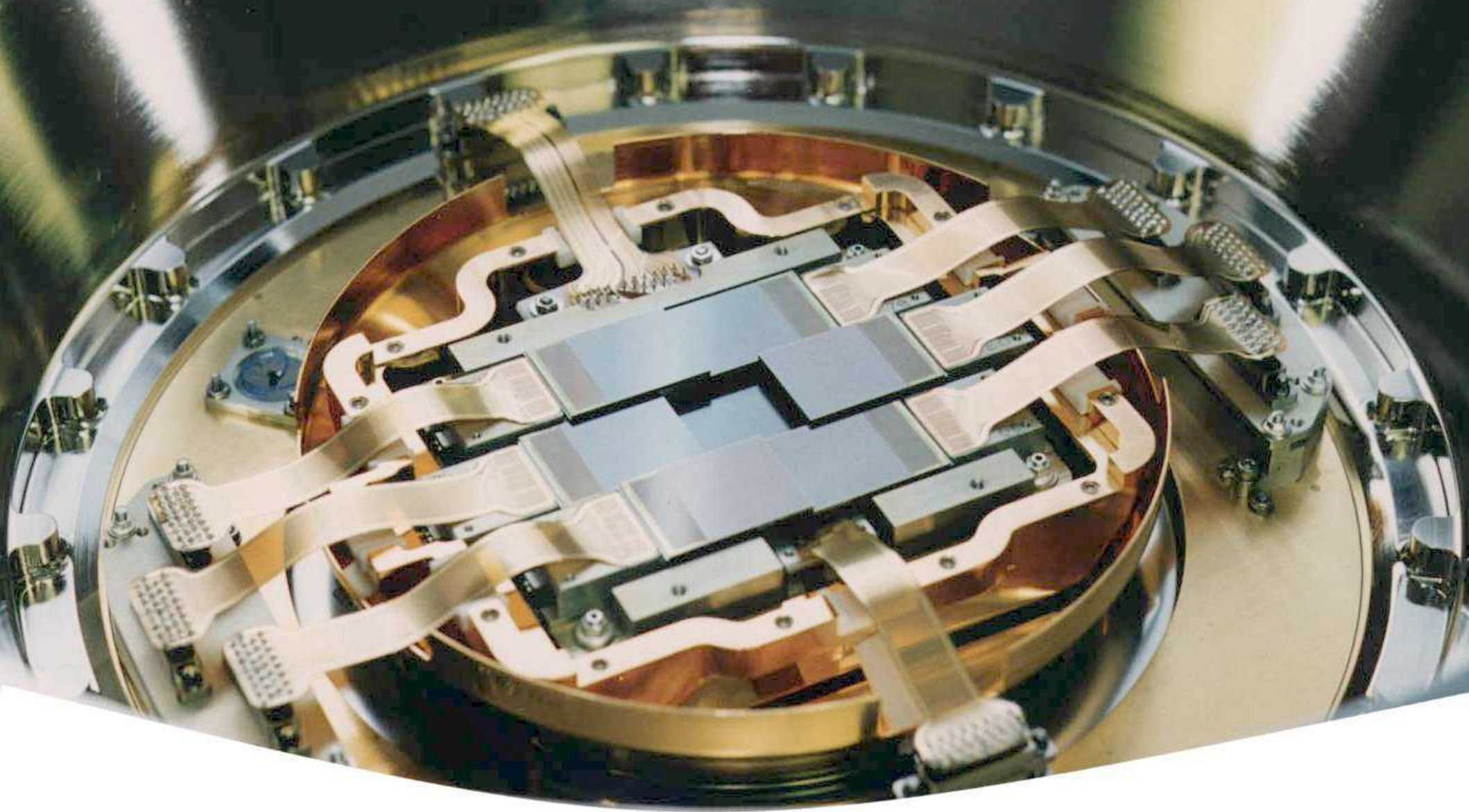
Camera's I have known and loved...

- Calibration scientist for the EPIC MOS cameras on XMM-Newton 2003-2007



Camera's I have known and loved...

- Calibration scientist for the EPIC MOS cameras on XMM-Newton 2003-2007



Course plan

Week	First half of evening	Break	Second half of evening
1	Introduction & basics	<i>Cafe is open 6pm-8:30pm!</i>	Basics inside PHP
2	Wide angle photography		Targets in the planetarium
3	Small 'scope observing (may change due to weather)		Small 'scope observing
4	Sun, Moon & planets		Processing Software
5	Zoom lenses & telescopes		Participants photo's
6	Observing with 28"		Video astronomy (time permitting)



Your cameras

Who...

- Doesn't have a camera?
- Has a mobile phone camera?
- Has a compact camera?
- Has a DSLR?



History

- CCD & CMOS sensors invented around 1970 – both are arrays of capacitors, each collecting charge proportional to their intensity.
- Up until this time, all previous devices (e.g. TV) had just one pixel
- First digital camera: Kodak, 0.01Mpx in 1975 with 23s read-out time & 4kg!
- 1979: first CCD on 1m Kitt Peak National Observatory (50x more sensitive than film)
- 1990: first commercial digital camera 1990 Dycam Model 1, and a year later, the Kodak DCS-100 (1.3Mp)



Mobile phone cameras



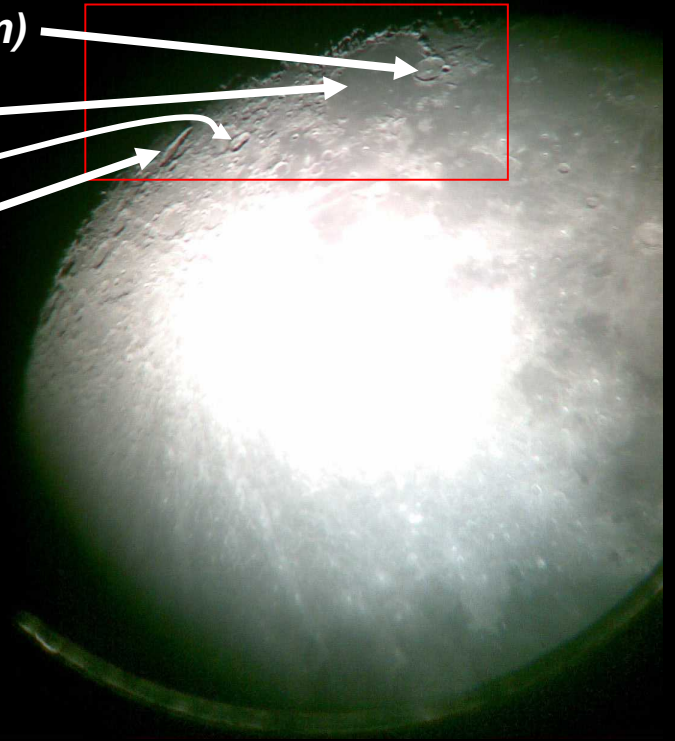


Schiller (180 km)

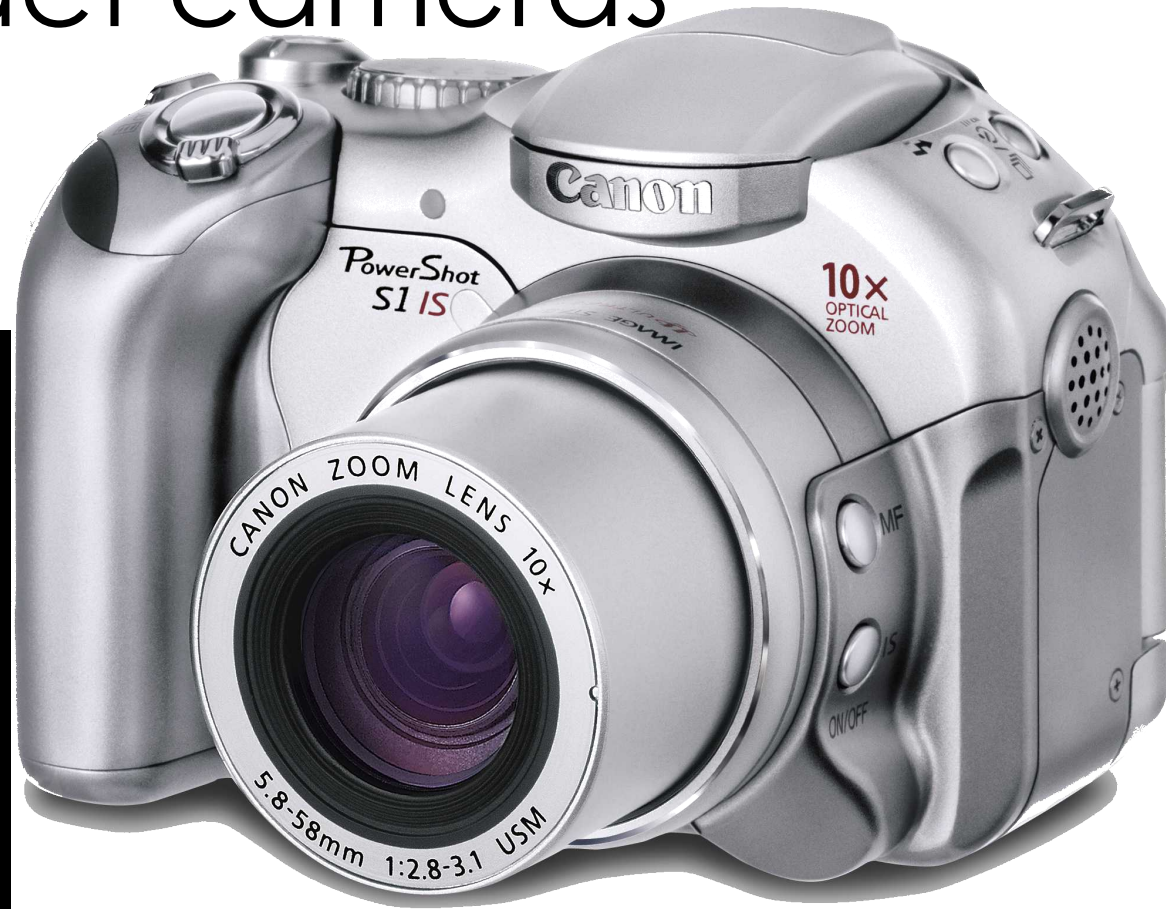
Hainzel (70 km)

Mare Humorum

Gassendi (110km)



Compact cameras













[clear](#)

dslr x

< \$2,000 x



low noise x

type

brand

general

lens

screen

movies

low light

image quality

features

shutter

storage

digicam

canon

price

zoom

large

1080p

image stabilization

high true resolution

raw

rapid fire

multiple slots

dslr

nikon

small

wide aperture

high-res

24p

large sensor

high megapixels

gps

no lag

high capacity

mirrorless

fujifilm

thin

wide angle

touch

slow motion

high iso

best overall quality

hdr

fast

compact flash

Found 3 cameras that meet your criteria, and 35 more that are close


[Specs](#) | [Prices](#) | [Videos](#)


#1 NIKON D600

£1,270.00 Pro DSLR, over a year ago

✓ DSLR ✓ £1,270.00

✓ Low noise up to 2,980 ISO

 Low noise at high ISO
2,980 ISO

 Great image quality
94.0

 Great color depth
25.1 bits

 Wide dynamic range
14.2 EV

 Large screen
3.2"

 Large viewfinder
0.70x

Great deal!

Buy now

WEX Photographic

£1,369


[Specs](#) | [Prices](#) | [Videos](#)


#2 NIKON D800

£1,829.99 Pro DSLR, over a year ago

✓ DSLR ✓ £1,829.99

✓ Low noise up to 2,853 ISO

Great deal!

Buy now

WEX Photographic

£1,962

[D800 vs the D600](#)

Advantages

- Much higher true resolution
- Higher resolution screen
- Less shutter lag

Tradeoffs

- Slightly more noise at high ISO
- Shoots slower
- Larger

clear

dslr x

< \$2,000 x

low noise x

type

brand

general

lens

screen

movies

low light

image quality

features

shutter

storage

digicam

canon

price

zoom

large

1080p

image stabilization

high true resolution

raw

rapid fire

multiple slots

dslr

nikon

small

wide aperture

high-res

24p

large sensor

high megapixels

gps

no lag

high capacity

mirrorless

fujifilm

thin

wide angle

touch

slow motion

high iso

best overall quality

hdr

fast

compact flash



Specs | Prices | Videos



#3 CANON EOS 6D

£1,257.00 Pro DSLR, over a year ago



DSLR



£1,257.00



Low noise up to 2,340 ISO

[Canon 6D vs the Nikon D600](#)

Advantages

- Better maximum light sensitivity
- Has a GPS
- Higher resolution screen

Tradeoffs

- Significantly worse image quality
- More noise at high ISO
- Significantly smaller screen

Great deal!

Buy now

WEX Photographic

£1,479



Specs | Prices | Videos



#4 CANON EOS 5D Mark III

£2,135.58 Pro DSLR, over a year ago



DSLR



£2,135.58



Low noise up to 2,293 ISO

[Canon 5D Mark III vs the Nikon D600](#)

Advantages

- Many more cross-type focus points
- Significantly less shutter lag
- Better maximum light sensitivity

Tradeoffs

- Significantly worse image quality
- Less dynamic range
- More noise at high ISO

Great deal!

Buy now

WEX Photographic

£2,329



#5 SONY

Camera settings

- Quality
- Exposure time
- ISO
- f-number



Quality



Quality







244 kb



46 kb



27 kb



16 kb

Exposure Time



Exposure Time



Exposure Time



Exposure Time



ISO Setting

- ISO is a standard measure of how sensitive the camera's sensor (or film) is to light
- Sensitivity can be increased by increasing the current in across the sensor...
- ...but this leads to more “noise” on the image

Sensitivity

Vs

Noise



ISO 100



ISO 200



ISO 400



ISO 800



ISO 1600



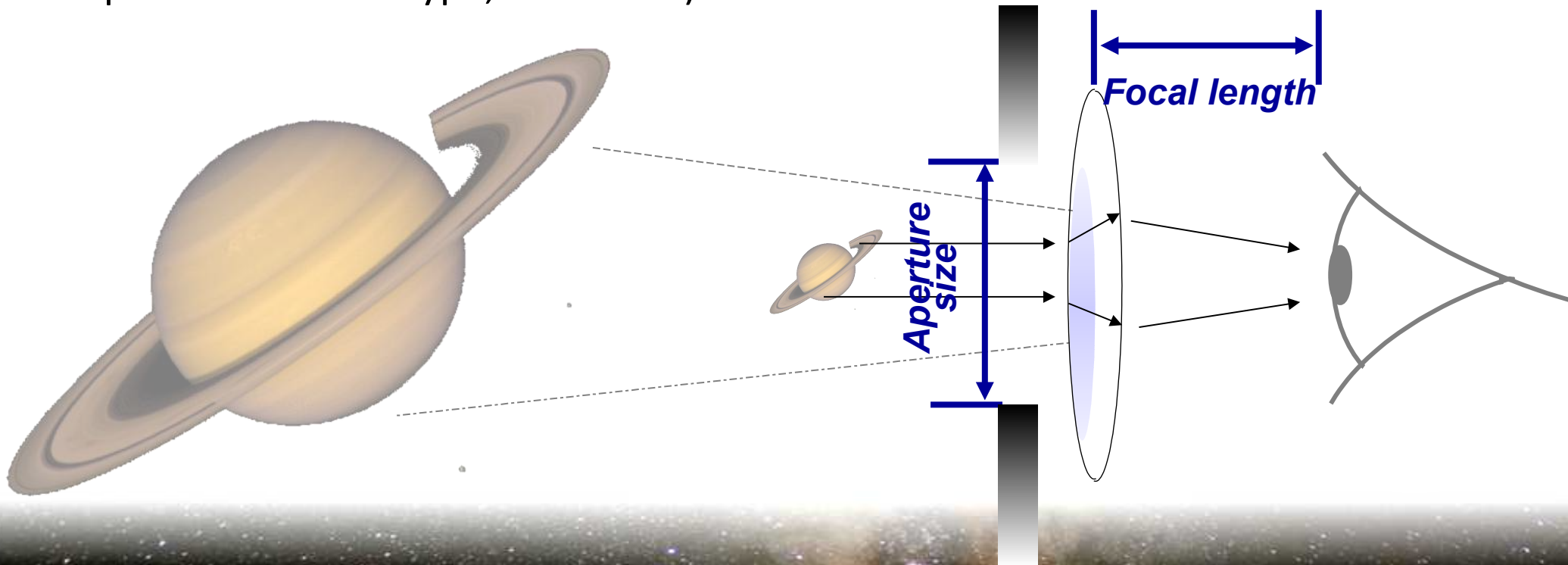
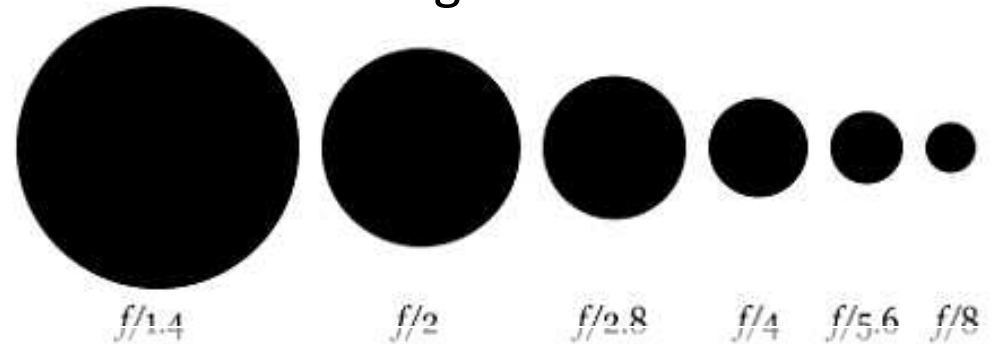
ISO 3200



f-number

Aperture size in terms of focal length of a lens

- e.g. $f/2$ means that the aperture size is half the focal length of the lens
- Affects amount of light getting into the camera
- Affects focal depth (which is also dependent on lens type, and zoom)



f/4.5



f/6.7



f/9.5



f/13



f/32



f/32



f/5



f-number

Amount of light collected

Vs

Depth of field/sharpness

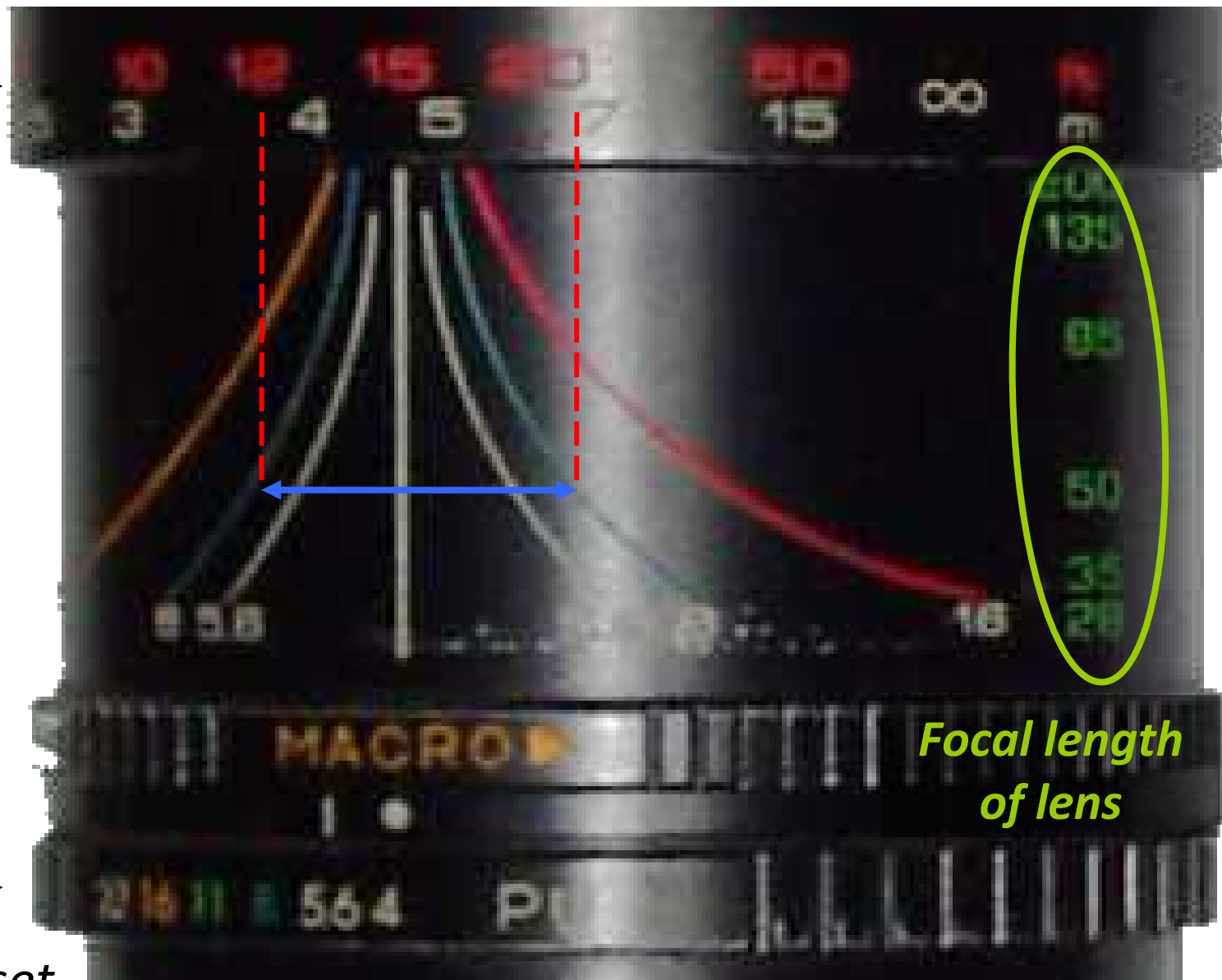




**Focus point
(ft & m)** →

**Range of the
focal point**

F number →



With this zoom lens set

to a mid-range 50mm focal length (neither wide-angle or zoomed), and aperture of f8, everything from 3.8m to 7m (12-22 feet) will be in focus.

f-number

Why is the f-number important in astrophotography?

1. It is notoriously difficult to focus on stars...

“going down a stop or two” (increasing the f-number) will help - but you will need a longer exposure to compensate

2. To be sharply focus on objects both near & far (foreground object & astronomical target behind)

3. Can give stars the classic “spikes” appearance





Settings Summary

<i>Increasing...</i>	<i>Advantage</i>	<i>Disadvantage</i>
Exposure time	Collect more light (makes fainter objects clearer)	Need accurate & strong tripod/mount to avoid shake/drift
ISO sensitivity	Signal is amplified (makes fainter objects brighter/clearer)	Increases noise on image
F-number	Easier to focus (leading to a sharper image)	Decreases amount of light getting through, so fainter detail is lost
Quality	Higher quality images	Larger file size, so fewer images can be stored on memory card

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www.flickr.com/baskill

